What follows are some examples of asset allocation in the context of the volatility strategy graphed in Figure 2.1, where the risk in the left-hand tail is especially pronounced and a risk-free asset pays 1%.

4.1. SAFETY FIRST

In the safety first utility framework, investors do exactly what the name suggests—they seek safety first. Roy's (1952) utility is very simple: it is just zero or one depending on whether a portfolio return is greater or less than a pre-determined level. If it is less than this level, a disaster results. If the return is greater than this level, the disaster is averted. Safety first minimizes the chance of a disaster; it is ideal for agents for whom meeting a liability is crucial. Not surprisingly, since safety first takes care of the downside first, asset allocation in the safety-first approach is very straightforward: you conservatively hold safe assets up until the pre-determined level is satisfied, and then you take on as much risk as you can after the safety level is met.

A related approach, formulated by Manski (1988) and Rostek (2010), is quantile utility maximization. In this approach, agents care about the worst outcomes that can happen with a given probability. Probability cutoffs of a probability density function are called quantiles. Quantiles are intuitive measures of investor pessimism. An asset owner might look at the worst outcome that can occur in 90% of all situations, for example. In this case, the relevant outcome is what happens at the 0.1 quantile (which is the first decile).[19] Another asset owner might look at the worst outcome that might happen 50% of the time, which is what happens at the 0.5 quantile (which is the median). In Manski-Rostek's utility, the quantile is a parameter choice, like risk aversion, and it is a measure of the investor's attitude to downside risk. Because the quantile parameter choice is an alternative measure of investors' downside risk aversion, the asset allocation from this framework is similar to loss aversion, which we now discuss.

4.2. LOSS AVERSION OR PROSPECT THEORY

Developed in a landmark paper by Kahneman and Tversky in 1979, loss aversion was originally created as an alternative to pure, rational expected utility. It is, however, a variant of expected utility, broadly defined, because it involves measuring risk by separable functions of (subjective) probabilities and utility functions of returns. Daniel Kahneman won the 2002 Nobel Prize for his work combining psychology and economics, especially for his work on prospect theory. His long-time collaborator, Amos Tversky, unfortunately did not live long enough to be awarded

[19] Value-at-risk that is so commonly used in risk management is a quantile measure. People usually pick the 0.01 or 0.05 quantiles.